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EXAMPLE 11.18

Consider two sound waves with wavelengths 5 *m* and 6 *m*. If these two waves propagate in a gas with velocity 330 ms^{-1}. Calculate the number of beats per second.

Solution

Given $\lambda_1 = 5m$ and $\lambda_2 = 6m$

Velocity of sound waves in a gas is $v = 330\ ms^{-1}$

The relation between wavelength and velocity is $v = \lambda f \Rightarrow f = \frac{v}{\lambda}$

The frequency corresponding to wavelength

λ_1 is $f_1 = \frac{v}{\lambda_1} = \frac{330}{5} = 66\,Hz$

The frequency corresponding to wavelength

λ_2 is $f_2 = \frac{v}{\lambda_2} = \frac{330}{6} = 55\,Hz$

The number of beats per second is

$|f_1 - f_2| = |66 - 55| = 11$ beats per sec

EXAMPLE 11.19

Two vibrating tuning forks produce waves whose equation is given by $y_1 = 5\sin(240\pi t)$ and $y_2 = 4\sin(244\pi t)$. Compute the number of beats per second.

Solution

Given $y_1 = 5\sin(240\pi t)$ and $y_2 = 4\sin(244\pi t)$

Comparing with $y = A\sin(2\pi f_1 t)$, we get

$2\pi f_1 = 240\pi \Rightarrow f_1 = 120Hz$

$2\pi f_2 = 244\pi \Rightarrow f_2 = 122Hz$

The number of beats produced is $|f_1 - f_2|$ $= |120 - 122| = |-2| = 2$ beats per sec

11.8 STANDING WAVES

11.8.1 Explanation of stationary waves

When the wave hits the rigid boundary it bounces back to the original medium and can interfere with the original waves. A pattern is formed, which are known as standing waves or stationary waves. Consider two harmonic progressive waves (formed by strings) that have the same amplitude and same velocity but move in opposite directions. Then the displacement of the first wave (incident wave) is

$$y_1 = A\sin(kx - \omega t) \quad (11.59)$$

(waves move toward right)

and the displacement of the second wave (reflected wave) is

$$y_2 = A\sin(kx + \omega t) \quad (11.60)$$

(waves move toward left)

both will interfere with each other by the principle of superposition, the net displacement is

$$y = y_1 + y_2 \quad (11.61)$$

Substituting equation (11.59) and equation (11.60) in equation (11.61), we get

$$y = A\sin(kx - \omega t) + A\sin(kx + \omega t) \quad (11.62)$$

Using trigonometric identity, we rewrite equation (11.62) as

$$y(x, t) = 2A\cos(\omega t)\sin(kx) \quad (11.63)$$